

TRDP II

BERT-SHAP Presentation Notes

A simple speech guide with equations and pronunciation

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This note explains what was done in the BERT-SHAP experiment, how each equation works, and how to pronounce every symbol while presenting. It is written as a speaking guide rather than as a formal research paper.

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1 The short version

A one-minute explanation

In our earlier experiment, we tried to explain BERT using an attention-based recursive feature elimination method, or RFEM. The method did not behave reliably. In some sentences, it removed words that clearly carried the sentiment. For example, a strong positive or negative word could be treated as unimportant.

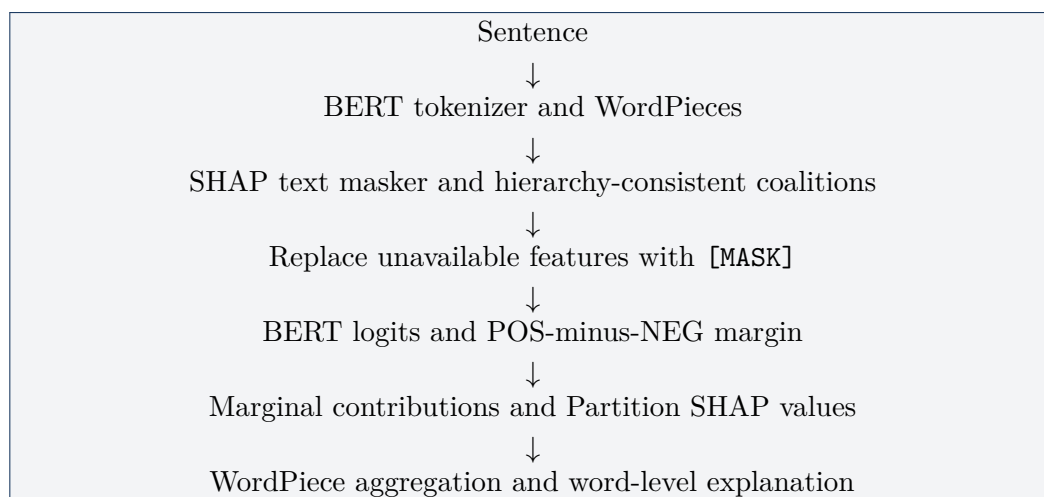
For the new experiment, I kept the same BERT sentiment classifier and changed the explanation method to SHAP. The central idea is simple: hide different words or groups of words, run the changed sentence through BERT, and observe how the model score changes. A word receives a positive SHAP value if it pushes the score toward positive sentiment and a negative SHAP value if it pushes the score toward negative sentiment.

Because checking every possible word combination would be too expensive, I used Partition SHAP. It groups related text features in a hierarchy and evaluates coalitions that follow that structure. BERT sometimes splits one word into several WordPieces, so I added their SHAP values together to recover one readable value for the original word.

The experiment used ten sentences. BERT predicted all ten labels correctly, and the saved SHAP values exactly reconstructed the explained logit margin at the stored precision. The explanations also identified meaningful words such as *beautiful*, *outstanding*, *boring*, *weak*, and *captivating*. Faithfulness tests showed that removing the most important words usually caused a large fall in the model's decision score.

Speaking tip: Do not try to recite every equation in the opening. First explain the idea in ordinary language. Introduce an equation only when the listener asks how a particular quantity was calculated.

The implemented flow



2 Model input and prediction

1. Tokenization

$$\tau(x) = (t_0, t_1, \dots, t_{M+1})$$

Meaning: x is the input sentence. The Greek letter τ , pronounced “tau,” represents the tokenizer. Each t_i is a token. M is the number of explainable features. In BERT, t_0 is normally [CLS] and t_{M+1} is [SEP].

What I would say: “The tokenizer converts the sentence into a sequence of BERT tokens. Some tokens are complete words, while others are WordPiece fragments.” **Say it aloud:** “Tau of x equals t sub zero, t sub one, and so on, up to t sub M plus one.”

2. BERT logits

$$\mathbf{z}_\theta(x) = (z_{\text{NEG}}(x), z_{\text{POS}}(x))$$

Meaning: $\mathbf{z}$, or “bold z ,” is the vector of raw model scores. θ , pronounced “theta,” represents the learned BERT parameters. The first score is for the negative class and the second is for the positive class. These raw scores are called logits.

What I would say: “BERT produces two raw scores: one for negative sentiment and one for positive sentiment.” **Say it aloud:** “Bold z sub θ of x equals z sub negative of x and z sub positive of x .”

3. Softmax probability

$$p_\theta(y = c \mid x) = \frac{\exp(z_c(x))}{\exp(z_{\text{NEG}}(x)) + \exp(z_{\text{POS}}(x))}$$

Meaning: p is probability, y is the output label, c is one chosen class, and the vertical bar means “given.” The notation $\exp$, read “exponential,” converts logits to positive values. Dividing by the sum makes the two probabilities add to one.

What I would say: “Softmax changes the two raw logits into probabilities. The class with the larger probability becomes the prediction.” **Say it aloud:** “ P sub θ of y equals c , given x , equals exponential of z sub c of x , divided by exponential of z sub negative of x plus exponential of z sub positive of x .”

4. Predicted class

$$\hat{y} = \arg \max_c p_\theta(y = c \mid x)$$

Meaning: $\hat{y}$, pronounced “ y hat,” is the predicted label. The term $\arg \max$, read “arg max,” means the class value c that produces the largest probability.

What I would say: “The prediction is simply whichever class has the highest probability.” **Say it aloud:** “ Y hat equals arg max over c of p sub θ of y equals c , given x .”

3 The score explained by SHAP

5. POS-minus-NEG logit margin

$$f(x) = z_{\text{POS}}(x) - z_{\text{NEG}}(x)$$

Meaning: $f(x)$ is the scalar score explained by SHAP. A positive value supports the positive class; a negative value supports the negative class. A larger magnitude means a stronger preference. The experiment explains logits rather than probabilities because logits do not saturate near zero or one.

What I would say: “Instead of explaining two outputs separately, I explain the positive logit minus the negative logit. This gives one signed decision score.” **Say it aloud:** “F of x equals z sub positive of x minus z sub negative of x.”

6. A coalition of available words

$$\mathbf{m} \in \{0, 1\}^M$$

Meaning: $\mathbf{m}$, or “bold m,” is a binary coalition vector. The symbol $\in$ means “belongs to.” If $m_i = 1$, feature i is available. If $m_i = 0$, it is unavailable. The superscript M means the vector contains one zero-or-one choice for each of the M explainable features.

What I would say: “Every coalition describes which text features are visible and which ones are hidden.” **Say it aloud:** “Bold m belongs to the set zero and one to the power M .”

7. Text masking rule

$$h_x(\mathbf{m})_i = \begin{cases} t_i, & m_i = 1, \\ [\text{MASK}], & m_i = 0. \end{cases}$$

Meaning: h_x is the reconstruction or masking function for sentence x . If a feature is available, its original token t_i remains. If it is unavailable, it is replaced by BERT’s [MASK] token. Special tokens such as [CLS] and [SEP] remain part of model processing.

What I would say: “SHAP creates a perturbed sentence by retaining available tokens and masking unavailable ones. Each perturbed sentence is then passed through the same unchanged BERT model.” **Say it aloud:** “H sub x of bold m, sub i, equals t sub i if m sub i equals one; otherwise it equals mask if m sub i equals zero.”

8. Value of one coalition

$$v_x(S) = f(h_x(\mathbf{1}_S))$$

Meaning: S is a subset of the available features. $\mathbf{1}_S$ is the indicator vector for that subset: it has ones for features in S and zeros elsewhere. The masking function builds the corresponding sentence, and f gives its BERT decision margin.

What I would say: “The value of a coalition is the BERT margin obtained after showing only the features selected by that coalition.” **Say it aloud:** “V sub x of S equals f of h sub x of the indicator vector of S.”

4 How SHAP assigns contributions

9. The Shapley value

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} \times [v_x(S \cup \{i\}) - v_x(S)]$$

Meaning: ϕ_i , pronounced “phi sub i,” is the attribution of feature i . The symbol $\sum$ means “sum.” The notation $S \subseteq N \setminus \{i\}$ means all subsets of feature set N that do not contain i . Vertical bars around S mean the size of the set. The exclamation mark means factorial. The symbol $\cup$ means union. The bracketed difference is the marginal contribution made when feature i is added to coalition S . The factorial fraction gives the correct Shapley weighting to coalition sizes.

What I would say: “A SHAP value is the weighted average change caused by adding one feature across many different contexts. This makes the attribution depend on what other words are present, not only on the word in isolation.” **Say it aloud:** “Phi sub i equals the sum over all subsets S of N excluding i , of S size factorial times M minus S size minus one factorial, divided by M factorial, multiplied by v sub x of S union i minus v sub x of S .”

Speaking tip: When presenting the Shapley equation, point first to the final bracket. That bracket contains the main idea: compare the model value before and after adding feature i . The terms before it provide the averaging weights.

10. Partition SHAP and Owen-style values

$$\phi_i^{\mathcal{T}} = \frac{1}{|\Pi_{\mathcal{T}}|} \sum_{\pi \in \Pi_{\mathcal{T}}} [v_x(P_i^{\pi} \cup \{i\}) - v_x(P_i^{\pi})]$$

Meaning: $\mathcal{T}$, read “calligraphic T,” is the feature hierarchy. $\Pi_{\mathcal{T}}$, read “capital pi sub calligraphic T,” is the set of permutations allowed by that hierarchy. Lowercase π , pronounced “pi,” is one such ordering. P_i^{π} is the set of features appearing before feature i in ordering π . The equation averages the marginal contribution over hierarchy-consistent contexts.

What I would say: “Exact Shapley values would require examining up to two to the power M coalitions. Partition SHAP reduces the cost by using a text feature hierarchy and estimating Owen-style values from coalitions that respect that hierarchy.” **Say it aloud:** “Phi sub i superscript calligraphic T equals one over the size of capital pi sub calligraphic T, times the sum over pi in capital pi sub calligraphic T, of v sub x of P sub i superscript pi union i , minus v sub x of P sub i superscript pi.”

11. Additive explanation model

$$g(\mathbf{m}) = \phi_0 + \sum_{i=1}^M \phi_i m_i$$

Meaning: g is the additive explanation. ϕ_0 , read “phi sub zero,” is the base value when no explainable content feature is available. Each included feature contributes ϕ_i , so the explanation is the base plus the sum of active feature contributions.

What I would say: “SHAP represents the local model score as a baseline plus one signed contribution from each visible feature.” **Say it aloud:** “G of bold m equals phi sub zero plus the sum from i equals one to M of phi sub i times m sub i.”

12. Local accuracy

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i$$

Meaning: For the original unmasked sentence, all feature indicators equal one. The base value plus every SHAP value should therefore reproduce the exact BERT margin $f(x)$. This is SHAP’s local accuracy, or additivity, property.

What I would say: “If I add the baseline and all token contributions, I should recover the model score for the full sentence.” **Say it aloud:** “F of x equals phi sub zero plus the sum from i equals one to M of phi sub i.”

13. Additivity residual

$$\epsilon_{\text{add}} = f(x) - \left(\phi_0 + \sum_{i=1}^M \phi_i \right)$$

Meaning: ϵ , pronounced “epsilon,” is the reconstruction error. The subscript “add” stands for additivity. A value of zero means the stored SHAP explanation reconstructs the stored model margin exactly. The maximum stored residual in this experiment was 0.0.

What I would say: “I verified that the explained model score and the sum of SHAP components agree. For all ten stored explanations, the residual was zero at the saved precision.” **Say it aloud:** “Epsilon sub add equals f of x minus the quantity phi sub zero plus the sum from i equals one to M of phi sub i.”

5 From BERT pieces to readable words

14. WordPiece aggregation

$$\Phi_w = \sum_{i \in I_w} \phi_i$$

Meaning: Φ_w , pronounced “capital phi sub w,” is the value of complete word w . I_w is the set of BERT token positions that belong to that word. The token-level values are added without changing their signs. For example, **res**, **##ili**, and **##ence** are joined and shown as *resilience*.

What I would say: “Because BERT can split one word into several WordPieces, I sum their SHAP values to obtain one understandable value for the original word.” **Say it aloud:** “Capital phi sub w equals the sum over i in I sub w of phi sub i.”

6 Faithfulness evaluation

15. Decision-direction score

$$d = \begin{cases} +1, & \hat{y} = \text{POS}, \\ -1, & \hat{y} = \text{NEG}, \end{cases} \quad q(x) = d f(x)$$

Meaning: d is the decision direction. It is plus one for a positive prediction and minus one for a negative prediction. Multiplying the margin by d creates $q(x)$, a score for support of the model’s own predicted class. A larger q therefore means stronger support, whichever class was predicted.

What I would say: “This sign conversion lets me evaluate positive and negative examples with the same rule. In both cases, a high q score supports the current decision.” **Say it aloud:** “D equals plus one if y hat equals positive and minus one if y hat equals negative; q of x equals d times f of x.”

16. Number of selected words

$$k_\alpha = \max(1, \lceil \alpha W \rceil)$$

Meaning: α , pronounced “alpha,” is the chosen fraction of words. W is the number of content-word groups. The ceiling marks $\lceil \cdot \rceil$ mean “round upward.” The maximum with one guarantees that at least one word is selected. R_α denotes the resulting top-ranked word set.

What I would say: “I choose the top alpha fraction of words according to how strongly their signed SHAP values support the predicted class.” **Say it aloud:** “K sub alpha equals the maximum of one and the ceiling of alpha times W.”

17. Comprehensiveness

$$C_\alpha = q(x) - q(x \setminus R_\alpha)$$

Meaning: $x \setminus R_\alpha$ means the sentence with the selected words made unavailable. Comprehensiveness measures how much the decision score falls after these supposedly important words are removed. A larger value is better because it shows that the selected words mattered to the original decision.

What I would say: “If the explanation selected genuinely important words, masking them should make the model much less confident in its original decision.” **Say it aloud:** “C sub alpha equals q of x minus q of x without R sub alpha.”

18. Sufficiency gap

$$S_\alpha = q(x) - q(x_{R_\alpha})$$

Meaning: x_{R_α} means a sentence representation in which only the selected words remain available. The sufficiency gap measures how much decision support is lost when the model receives only those words. A smaller gap is better. A negative gap is possible if the selected subset supports the decision more strongly than the complete sentence.

What I would say: “The selected words are sufficient when keeping only them preserves most of the original decision score.” **Say it aloud:** “S sub alpha equals q of x minus q of x sub R sub alpha.”

19. Cross-sentence word importance

$$G(w) = \frac{1}{n_w} \sum_{j=1}^{n_w} |\Phi_{w,j}|$$

Meaning: n_w is the number of times word w appears in the ten sentences. $\Phi_{w,j}$ is its word-level SHAP value at occurrence j . Absolute value bars remove the direction so that strong positive and strong negative evidence both count as important. The result is a descriptive mean absolute value for this small sentence set, not a population-level global explanation.

What I would say: “For a cross-sentence view, I average the absolute word-level SHAP values. This measures strength, regardless of whether the word pushes positive or negative.”

Say it aloud: “G of w equals one over n sub w times the sum from j equals one to n sub w of the absolute value of capital phi sub w comma j.”

7 What the results mean

Main numerical results

Sentences evaluated	10
Correct predictions	10 out of 10
Maximum stored additivity residual	0.0
Mean comprehensiveness drop	5.3107; higher is better
Mean sufficiency gap	0.8927; lower is better

Examples that are easy to explain aloud

- In S1, *beautiful* has a SHAP value of (+2.2569). It strongly pushes the margin toward positive sentiment.
- In S3, *outstanding* contributes (+3.7479), while *slow* contributes (-1.0241). SHAP captures both sides of the contrast.
- In S7, *boring* contributes (-3.5674), which strongly supports the negative prediction.
- In S8, *captivating* contributes (+4.7673), while *weak* contributes (-3.8786). This is the clearest example of opposing evidence within one sentence.

A simple results speech

“I tested the implementation on the same ten qualitative sentences. BERT predicted every gold label correctly, although this small set is not a proper accuracy benchmark. The stored SHAP decompositions also passed the additivity check: the baseline plus all feature contributions reconstructed the explained logit margin with a maximum stored residual of zero.

The word explanations were more meaningful than the earlier RFEM results. In a positive example, *beautiful* received a strong positive value. In a negative example, *boring* received a strong negative value. In mixed sentences, SHAP represented competing evidence: *outstanding* opposed *slow*, and *captivating* opposed *weak*.

The mean comprehensiveness drop was 5.3107. This means the decision score fell substantially when the highest-ranked words were masked. The mean sufficiency gap was 0.8927. This means that retaining only the highest-ranked words preserved much of the original decision support. Together, these results suggest that the selected words are relevant to the model’s behavior.”

8 Limitations to mention

1. **SHAP explains the model, not objective language truth.** A value describes how this BERT classifier behaved under this explanation setup.
2. **The values are context-dependent.** A word can receive a different attribution in another sentence because its contribution depends on surrounding words.
3. **Masking is an artificial intervention.** Replacing words with [MASK] may create inputs that differ from naturally written sentences.

4. **Partition SHAP is structured and approximate.** It uses a feature hierarchy instead of evaluating every one of the 2^M coalitions.
5. **The sample is small.** Ten sentences are useful for qualitative comparison with RFEM, but they do not establish general performance on an unseen test set.
6. **Additivity is a consistency check, not complete validation.** A zero residual proves that the stored components reconstruct the chosen score; it does not prove that [MASK] is a perfect linguistic baseline.

A balanced conclusion

“The main improvement over RFEM is that SHAP measures the effect of hiding features and averaging their marginal contributions across multiple contexts. In these examples, it assigns substantial contributions to recognizable sentiment words and represents positive and negative evidence in the same sentence. The faithfulness tests support those selections. However, the explanations remain local to this BERT model, these sentences, the POS-minus-NEG margin, and the chosen masking method.”

9 Pronunciation quick reference

Symbol	Pronunciation	Meaning
τ	tau, rhymes with “cow”	tokenizer function
θ	theta	learned model parameters
ϕ	phi, commonly “fie”	token-level SHAP value
Φ	capital phi	aggregated word-level SHAP value
ϵ	epsilon	reconstruction error
α	alpha	selected fraction of words
π	pi	one feature ordering or permutation
Π	capital pi	a set of allowed permutations
$\mathcal{T}$	calligraphic T	the Partition SHAP hierarchy
$\hat{y}$	y hat	predicted class
z_i	z sub i	z with subscript i
P_i^π	P sub i superscript pi	predecessors of i in ordering pi
$\in$	belongs to	set membership
$\subseteq$	is a subset of	subset relationship
$\setminus$	without, or set minus	elements removed from a set
$\cup$	union	combine two sets
$\sum$	sum	add a sequence of terms
$\arg \max$	arg max	input or class that gives the maximum
$\exp(z)$	exponential of z	the exponential function
$ S $	size of S	number of elements in a set
$ a $	absolute value of a	magnitude without sign
$M!$	M factorial	$M(M-1)\cdots 1$
$\lceil a \rceil$	ceiling of a	round upward to an integer
$y = c \mid x$	y equals c, given x	conditional statement

10 Final two-minute speech

“Our earlier RFEM experiment did not always keep the most meaningful sentiment words, so I implemented SHAP as a second explanation method while keeping the same BERT classifier and qualitative sentences.

The sentence is first converted into BERT tokens. SHAP then creates coalitions, which are different combinations of available and unavailable features. An unavailable feature is replaced with the BERT [MASK] token, and every perturbed sentence is passed through BERT. The scalar score I explain is the positive logit minus the negative logit. Therefore, a positive SHAP value pushes the prediction toward positive sentiment, and a negative value pushes it toward negative sentiment.

The key Shapley idea is to compare the model value before and after one feature is added, then average that marginal contribution across different contexts. Partition SHAP makes this feasible for text by following a hierarchy of feature groups instead of checking all two-to-the-power-M coalitions. After the token-level values are calculated, I sum the values of WordPieces that belong to the same original word.

I checked the explanation arithmetic using the additivity residual. The BERT margin should equal the SHAP base value plus all feature contributions. The maximum stored residual was zero. I also evaluated faithfulness. The mean comprehensiveness drop was 5.3107, meaning that masking the selected words caused a large reduction in decision support. The mean sufficiency gap was 0.8927, meaning that the selected words retained much of that support when shown alone.

The local examples were intuitive. *Beautiful*, *outstanding*, and *captivating* received strong positive contributions, while *boring* and *weak* received strong negative contributions. This is more useful than the earlier RFEM behavior because the important sentiment cues remain visible in the explanation.

My conclusion is that Partition SHAP provides a clearer local explanation of this BERT classifier for these examples. However, the values are not permanent dictionary meanings. They depend on the sentence, the model, the output score, and the way unavailable words are masked.”

One sentence to remember

SHAP explains a prediction by measuring how the model score changes when a feature is added to or removed from different feature coalitions.